

TEMPLE UNIVERSITY
FOX SCHOOL OF BUSINESS

AS 3502 / AS 5103 Advanced Long-Term Actuarial Modeling
FALL 2026

HOMEWORK #1

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Due Friday, September 11, 2026 2:00pm

Show all work and circle your final answers.

1. You are given the following matrix containing annual Markov transition probabilities with states 0, 1, 2 for (x). State (0) represents Healthy, state (1) represents Disabled, and state (2) represents Dead.

$$\begin{bmatrix} 0.85 & 0.11 & 0.04 \\ 0.10 & 0.84 & 0.06 \\ 0 & 0 & 1 \end{bmatrix}$$

- (a) Calculate ${}_3p_x^{12}$.
- (b) There are six Healthy lives age (x) with independent future lifetimes. Calculate the probability that exactly three of them will be Disabled at the end of 2 years.

2. You are given a three-state model about (x)

State 0: Healthy
State 1: Disabled
State 2: Dead

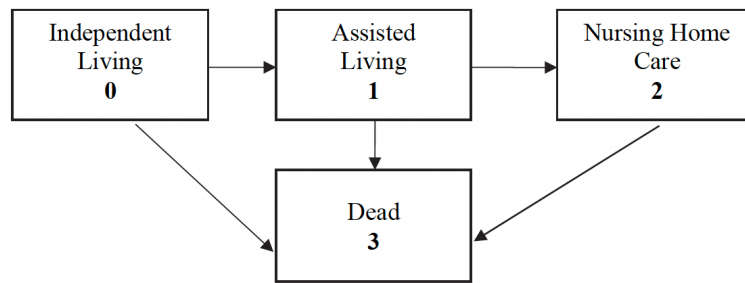
You are given the following transition intensities for all t:

- (i) $\mu_x^{01}(t) = 0.03$; $\mu_x^{02}(t) = 0.026$; $\mu_x^{12}(t) = 0.05$
(ii) All other transition intensities are equal to 0.

Create a transition matrix containing values of ${}_6p_x^{ij}$.

Calculate each transition probability **independently** and then check that each row in the matrix sums to 1. Do not solve any probabilities by summing up other values in the row and subtracting that sum from 1.

3. You are given the following Markov model:

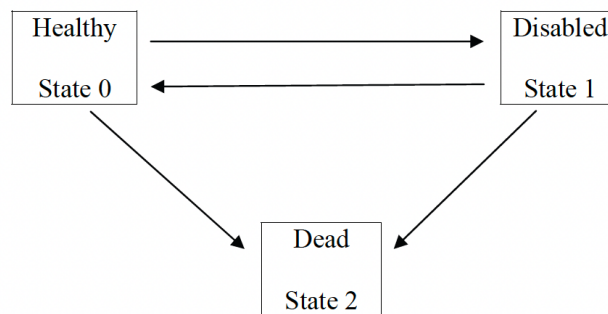


The forces of transition are the following:

- $\mu^{01} = 0.01$
- $\mu^{03} = 0.02$
- $\mu^{12} = 0.30$
- $\mu^{13} = 0.40$
- $\mu^{23} = 0.70$

Calculate the probability that a person currently in Independent Living will be in Assisted Living at the end of 3.5 years.

4. You are given the following Markov model:



You assume the following constant forces of transition:

- (i) $\mu^{01} = 0.06$
- (ii) $\mu^{10} = 0.03$
- (iii) $\mu^{02} = 0.01$
- (iv) $\mu^{12} = 0.04$

- (a) Write the Kolmogorov forward differential equations with the associated boundary conditions for this model for:

(i) ${}_t p_x^{11}$

(ii) ${}_t p_x^{10}$

(iii) ${}_t p_x^{12}$

- (b) (x) is currently in state 1. Using Euler's method and Kolmogorov's forward equations with step $h = 1/4$, calculate the probability that (x) will be in state 0 at the end of six months.